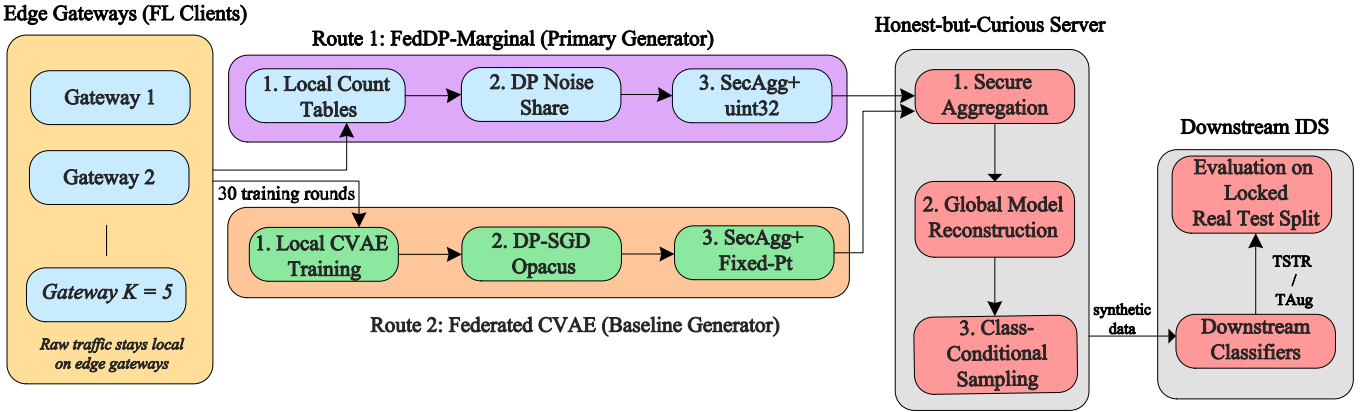

Edge Gateways (FL Clients)
Gateway 1
Gateway 2
Gateway K = 5
Raw traffic stays local on edge gateways
Route 1: FedDP-Marginal (Primary Generator)
1. Local Count Tables
2. DP Noise Share
3. SecAgg+ uint32
30 training rounds
1. Local CVAE Training
2. DP-SGD Opacus
3. SecAgg+ Fixed-Pt
Route 2: Federated CVAE (Baseline Generator)
Honest-but-Curious Server
1. Secure Aggregation
2. Global Model Reconstruction
3. Class-Conditional Sampling
synthetic data
Downstream IDS
Evaluation on Locked Real Test Split
TSTR / TAug
Downstream Classifiers